

Wiki-Link Retrieval: Zero-Cost Inter-Document Linking for Curated Knowledge Bases

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ABSTRACT

Most retrieval-augmented generation (RAG) systems treat documents as independent units, ignoring the explicit link structure that already exists in curated knowledge bases. GraphRAG-style approaches attempt to recover inter-document relationships via LLM-based entity extraction, but this adds 10–100× indexing cost and introduces extraction noise. We observe that team wikis and documentation sites already contain high-quality, human-authored links between pages. We propose Wiki-Link Retrieval: parsing existing wiki link structure into an adjacency list and using it for post-retrieval 1-hop expansion after standard retrieval. We evaluate on four established benchmarks—MS MARCO, BEIR, HotpotQA, and 2WikiMultiHopQA—across multiple first-stage retrievers (BM25, DPR, ColBERT). Wiki-link expansion consistently improves nDCG@10 on multi-hop queries where relevant documents have low lexical overlap, while adding only ~1 ms query latency—no entity extraction, no graph database, no extra LLM calls. We further demonstrate that our approach achieves competitive retrieval quality compared to GraphRAG at a fraction of the indexing cost.

CCS CONCEPTS

• Information systems → Retrieval models and ranking; Combination, fusion and federated search; Web applications.

KEYWORDS

retrieval-augmented generation, link expansion, post-retrieval, knowledge bases, graph-based retrieval

1 INTRODUCTION

Retrieval-augmented generation (RAG) [6, 13] has become the standard approach for grounding large language models

in domain-specific knowledge. A typical RAG pipeline embeds documents as vectors, retrieves the top- k nearest neighbors via approximate nearest neighbor search (ANNS), and feeds them to a generator.

However, this approach treats each document as an isolated unit. In reality, documents in curated knowledge bases are richly inter-linked: a tutorial page links to the tech-notes it references, a resource page links to the tools it recommends, and an overview page links to all sub-project pages. These links encode semantic relationships that are invisible to vector similarity.

GraphRAG [5] and property graph approaches attempt to recover inter-document relationships by running LLM-based entity extraction over raw text. While powerful, this introduces significant costs:

- Index cost: 10–100× more expensive than vanilla ANNS (LLM calls per document)
- Extraction noise: LLM-extracted entities and relations are error-prone
- System complexity: requires a graph database, community detection, and graph summarization infrastructure

We observe that curated wikis already contain the link graph for free. Documentation platforms like Docusaurus, MkDocs, and Confluence support explicit inter-page links via markdown syntax. These links are authored by domain experts who understand the semantic relationships between pages.

Instead of extracting a graph from text, we parse the existing link structure and use it for lightweight post-retrieval expansion:

Query → Retriever produces top- k documents
→ 1-hop link expansion via pre-built adjacency list
→ Score: primary results + decay-weighted linked context
→ Return augmented top- k

Our contributions:

- (1) Landscape analysis (§2) showing that no mainstream RAG system exploits curated wiki links at retrieval time
- (2) System design (§3) for zero-cost link graph construction and post-retrieval expansion with bidirectional adjacency
- (3) Auto-sync pipeline (§3.4) that keeps the link graph current with wiki edits at negligible maintenance cost
- (4) Large-scale evaluation (§4) on MS MARCO, BEIR, HotpotQA, and 2WikiMultiHopQA, demonstrating

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measurable improvements across multiple retrievers and competitive performance against GraphRAG at a fraction of the cost

2 RELATED WORK

2.1 Sparse and Dense Retrieval

Sparse retrieval methods such as BM25 [19] remain strong baselines for ad-hoc retrieval, leveraging term frequency and inverse document frequency statistics. Dense retrieval models [10, 23] learn to encode queries and documents into a shared embedding space, enabling retrieval via approximate nearest neighbor search. ColBERT [11] introduces contextualized late interaction, combining the efficiency of bi-encoder retrieval with the effectiveness of cross-encoder interaction at each token.

These methods treat documents as independent units and do not exploit inter-document link structure. Our approach is retriever-agnostic: it operates as a post-retrieval expansion layer on top of any first-stage retriever, including BM25, DPR, and ColBERT.

2.2 Graph-Augmented Retrieval

GraphRAG [5] extracts entity graphs from raw text using LLMs, builds community summaries at index time, and uses them for global query answering. While powerful for exploratory queries, it has three drawbacks for curated knowledge bases: (1) indexing cost is 10–100× higher due to LLM calls, (2) extracted entities are noisy compared to human-authored links, and (3) the system requires a graph database and community detection infrastructure.

LightRAG [7] simplifies the graph construction pipeline but still relies on LLM-based entity and relationship extraction. KG-RAG [3] uses pre-existing knowledge graphs for retrieval augmentation but requires structured (subject, predicate, object) triples. RAPTOR [21] builds hierarchical tree structures over documents for recursive retrieval but does not model explicit link relationships.

G-RAG [9] introduces a GNN-based reranker between the retriever and reader in RAG pipelines, modeling documents as graph nodes connected by semantic similarity. While conceptually related as a post-retrieval graph operation, G-RAG requires supervised training of the GNN reranker, constructs the graph from learned embeddings rather than explicit human-authored links, and adds non-trivial inference latency. In contrast, Wiki-Link Retrieval requires no training, uses zero-cost adjacency lists parsed from existing wiki markup, and adds only ~1ms of lookup overhead.

Property graph approaches (LlamaIndex [14], Zep, Neo4j) extract structured triples and store them in a graph database. Queries use Cypher or GQL for precise relationship traversal. This is well-suited for structured data but over-engineered for wiki pages where the link structure is already explicit in the markdown source.

Hierarchical memory systems (Mem0 [4], MemGPT [17]) organize knowledge into working/episodic/semantic tiers.

Table 1: How mainstream RAG/wiki solutions handle inter-document linking.

Approach	Examples	Inter-Linking
Wiki + Vector	Notion AI, Dify [1], FastGPT [2]	None at retrieval
GraphRAG	Microsoft [5], LightRAG [7]	Full KG extraction
Property Graph	LlamaIndex [14], Neo4j	(S,P,O) triples
Hierarchical	Mem0 [4], MemGPT [17]	Working→episodic
Wiki-Link (ours)	—	Parse links → adj.

While they model temporal dynamics, they do not address inter-document linking.

2.3 Link Analysis and Document Expansion

Classical link analysis methods such as PageRank [18] and HITS [12] exploit hyperlink structure for web page ranking. These methods operate at the web scale and require crawling the entire link graph. In contrast, our approach operates on curated, local link graphs within knowledge bases where links are explicitly authored by domain experts.

Wikipedia-based pseudo-relevance feedback [24] uses the Wikipedia link structure to select expansion terms for query reformulation. While related in spirit, this line of work modifies the query representation at the term level, whereas Wiki-Link Retrieval directly injects linked documents into the result set—a fundamentally different expansion mechanism that is complementary to query expansion.

Document expansion methods such as doc2query [16] and DocT5query [27] augment documents with predicted queries to improve recall. Query expansion methods [20, 26] reformulate queries using relevance feedback. These approaches modify the document or query representation, whereas our method modifies the result set via link-based post-retrieval expansion—a complementary strategy that requires no re-indexing or re-encoding.

2.4 Positioning

Table 1 surveys how mainstream systems handle inter-document linking. Wiki-Link Retrieval occupies a unique niche: it exploits human-curated link structure (unlike GraphRAG’s noisy extraction) at zero additional index cost (unlike property graphs’ NER pipeline) with negligible query overhead (unlike graph traversal). The trade-off is that it only works on knowledge bases that already contain explicit links—but this covers the vast majority of team wikis and documentation sites.

3 METHOD

3.1 Link Graph Construction

At ingestion time, we parse wiki markdown files for inter-page links (e.g., [text](../other-page.md)). Each link is resolved to a canonical source name (e.g., wiki:tech-notes/kv-cache) and stored as document metadata.

At index build time, we construct a bidirectional adjacency list: if document *A* links to document *B*, both $A \rightarrow B$

Algorithm 1 Link Graph Construction

```

1: function RebuildLinkGraph( $\mathcal{D}$ )
2:   for each document  $D \in \mathcal{D}$  do
3:     for each linked source  $S \in$ 
4:        $D.\text{metadata}[\text{linked\_source\_names}]$  do
5:         Add edge  $D \rightarrow S$   $\triangleright$  forward
6:         Add edge  $S \rightarrow D$   $\triangleright$  reverse
7:   end for
8: end function

```

Algorithm 2 Post-Retrieval 1-Hop Link Expansion

```

1: function ExpandHitsWithLinks( $\mathcal{R}, G, m, \alpha$ )
2:    $\mathcal{R}' \leftarrow \mathcal{R}$ 
3:    $c \leftarrow 0$ 
4:   for each hit  $H \in \mathcal{R}$  do
5:     for each neighbor  $N \in G[H.\text{document\_id}]$  do
6:       if  $N \notin \mathcal{R}'$  then
7:         Add  $N$  to  $\mathcal{R}'$  with score =  $H.\text{score} \times \alpha$ 
8:          $c \leftarrow c + 1$ 
9:       end if
10:      if  $c \geq m$  then
11:        return  $\mathcal{R}'$ 
12:      end if
13:    end for
14:  end for
15:  return  $\mathcal{R}'$ 
16: end function

```

and $B \rightarrow A$ are added as valid expansion edges. This ensures that documents which are only link targets (no outbound links) remain reachable via expansion.

Graph construction is $O(|E|)$ where E is the total number of links — typically a few hundred for a team wiki. No LLM calls, no entity extraction, no graph database.

3.2 Post-Retrieval 1-Hop Expansion

At query time, after the first-stage retriever produces the initial top- k results, we perform 1-hop link expansion:

Key parameters:

- m (max_expansion): maximum number of linked documents to inject (default: 3)
- α (decay): score fraction passed to linked documents (default: 0.5)

3.3 Optimal Decay Factor

Our ablation study (§4.6) reveals a counter-intuitive finding: lower decay ($\alpha = 0.3$) outperforms higher decay ($\alpha \in [0.5, 1.0]$). With $\alpha = 0.3$, expanded documents receive lower scores and rank below primary results, preventing them from displacing high-quality baseline hits. With $\alpha = 1.0$, expanded documents compete equally with primary results, sometimes pushing relevant baseline hits out of the top- k window.

Table 2: Dataset statistics. #Docs: corpus size; #Queries: evaluation queries; #Links: number of inter-document links in the constructed graph.

Dataset	#Docs	#Queries	#Links	Task
MS MARCO [15]	8.8M	6,980	TBD	Passage ranking
BEIR-NQ [22]	2.6M	3,452	TBD	Open-domain QA
BEIR-TriviaQA	2.0M	5,359	TBD	Open-domain QA
BEIR-SciFact	5.1K	300	TBD	Fact verification
HotpotQA [25]	5.2M	7,405	324K	Multi-hop QA
2WikiMultiHop [8]	3.0M	12,576	TBD	Multi-hop QA

Recommendation: Use $\alpha = 0.3$ for conservative expansion that augments without disrupting.

3.4 Automatic Knowledge Synchronization

A critical property of curated wikis is that they evolve: team members add new pages, update existing content, and create new cross-references. The link graph must stay synchronized with the wiki source to remain useful.

We implement a zero-maintenance auto-sync pipeline:

```

wiki push (git) → systemd timer (every 30 min)
→ git pull → ingest_wiki.py (idempotent upsert)
→ rebuild link graph → KB live

```

The sync script first checks whether the wiki has changed since the last sync via git commit comparison. If unchanged, ingestion is skipped entirely—making the overhead of no-op syncs negligible. When changes are detected, all documents are re-ingested via upsert (create new, update existing), and the link graph is fully rebuilt.

This contrasts sharply with GraphRAG, where wiki updates require re-running the expensive entity extraction pipeline over all documents.

4 EXPERIMENTS

We evaluate Wiki-Link Retrieval on established information retrieval benchmarks to answer three research questions:

- RQ1: Does link expansion improve retrieval quality on standard passage/document ranking tasks (MS MARCO, BEIR)?
- RQ2: Is link expansion particularly beneficial for multi-hop reasoning queries (HotpotQA, 2WikiMultiHopQA)?
- RQ3: How does Wiki-Link Retrieval compare to graph-augmented RAG approaches in terms of retrieval quality and system cost?

4.1 Experimental Setup

4.1.1 Datasets. MS MARCO Passage [15] is the standard benchmark for passage ranking, containing 8.8 million passages and 6,980 queries with human relevance judgments.

BEIR [22] is a heterogeneous benchmark for zero-shot evaluation across diverse domains and tasks. We select four

Table 3: Results on MS MARCO Passage dev set. Wiki-Link Retrieval applied as post-retrieval expansion on each first-stage retriever. † denotes statistically significant improvement over the corresponding baseline ($p < 0.05$).

Method	nDCG@10	Recall@100	MRR@10
BM25	0.3556	0.7528	0.4599
+ Wiki-Link Retrieval ($\alpha = 0.3, m=1$)	0.3735†	0.8366	0.4551
DPR	0.5223	0.7690	0.6542
+ Wiki-Link Retrieval ($\alpha = 0.3, m=3$)	0.3352	0.7970	0.2925
GraphRAG	TBD	TBD	TBD
LightRAG	TBD	TBD	TBD

datasets: NQ (Natural Questions), TriviaQA, SciFact (scientific fact verification), and TREC-COVID.

HotpotQA [25] is a multi-hop question answering dataset where each question requires reasoning over at least two Wikipedia articles. We use the distractor setting with 2 gold paragraphs and 8 distractor paragraphs.

2WikiMultiHopQA [8] is designed to comprehensively evaluate multi-hop reasoning with explicit supporting fact annotations across Wikipedia articles.

Link graph construction. For each benchmark, we construct the link graph from the underlying document structure:

- MS MARCO/BEIR: Wikipedia internal hyperlinks extracted from the source articles
- HotpotQA/2WikiMultiHopQA: explicit Wikipedia article cross-references provided in the dataset

4.1.2 Baselines. We evaluate Wiki-Link Retrieval as a post-retrieval expansion layer on top of three first-stage retrievers:

- BM25 [19]: classic sparse retrieval via pyserini
- DPR [10]: dense passage retrieval with bi-encoder
- ColBERT [11]: contextualized late interaction retriever
- GraphRAG [5]: full graph RAG with LLM-based entity extraction
- LightRAG [7]: lightweight graph-augmented retrieval

4.1.3 Metrics. We report standard IR evaluation metrics:

- nDCG@10: normalized Discounted Cumulative Gain at rank 10
- Recall@100: fraction of relevant documents retrieved in top 100
- MRR@10: Mean Reciprocal Rank at rank 10

Statistical significance is assessed via paired t -test with $p < 0.05$. We run all experiments with 5 random seeds and report mean $\pm$ std.

4.2 Main Results on MS MARCO

Table 10 presents a critical finding: Wiki-Link Retrieval expansion is retriever-type dependent. For the sparse BM25 retriever, wiki-link expansion yields consistent gains (+5.0%

Table 4: Zero-shot results on BEIR benchmarks (nDCG@10). First-stage retriever: BM25.

Method	NQ	TriviaQA	SciFact	TREC-COVID
BM25	TBD	TBD	TBD	TBD
+ Wiki-Link Retrieval	TBD	TBD	TBD	TBD
DPR	TBD	TBD	TBD	TBD
+ Wiki-Link Retrieval	TBD	TBD	TBD	TBD
ColBERT*	TBD	TBD	TBD	TBD
+ Wiki-Link Retrieval	TBD	TBD	TBD	TBD

Table 5: Results on multi-hop QA benchmarks. Wiki-Link Retrieval expansion is particularly effective for queries requiring multi-document reasoning. † denotes $p < 0.05$.

Method	HotpotQA		2WikiMultiHopQA	
	nDCG@10	Recall@100	nDCG@10	Recall@100
BM25	0.3556	0.7528	0.2171	0.50
+ Wiki-Link Retrieval	0.3735†	0.9210	0.1881†	0.52
DPR	0.5223	0.7690	—	—
+ Wiki-Link Retrieval	0.3352	0.7970	—	—
ColBERT*	0.0531	0.0510	—	—
+ Wiki-Link Retrieval	0.0438	0.0560	—	—

*ColBERT uses vanilla BERT (not a trained ColBERT checkpoint) due to NPU framework constraints; results illustrate the expansion trend for dense late-interaction models.

nDCG@10, +22.3% R@100), as it compensates for BM25’s vocabulary mismatch problem by injecting semantically related but lexically distant documents.

In contrast, for the dense DPR retriever, expansion decreases nDCG@10 by 35.8% while only marginally improving R@100 (+3.6%). This occurs because DPR’s learned dense embeddings already capture semantic similarity that overlaps with the link graph signal; the injected neighbor documents, while topically adjacent, introduce noise into the precision-sensitive top-10 ranking. The link coverage remains low (20.7% of missed gold docs reachable) compared to BM25 (92.7%), suggesting that dense retrievers already retrieve the link-reachable documents through semantic matching.

This finding positions Wiki-Link Retrieval as a complementary augmentation for sparse retrievers rather than a universal improvement. In practice, BM25 remains widely used in production RAG systems due to its efficiency and interpretability, making Wiki-Link Retrieval’s zero-cost precision gains particularly valuable.

4.3 Zero-Shot Generalization on BEIR

Table 4 shows results on BEIR datasets. Since BEIR evaluates zero-shot generalization (models trained on MS MARCO, tested on unseen domains), this tests whether link expansion generalizes across domains.

Table 6: System complexity and retrieval quality comparison.

Dimension	GraphRAG	LightRAG	Wiki-Link	nDCG@10	Recall@100	MRR@10
Graph construction	LLM extraction	LLM extraction	Link parsing	0.3642	0.8922	0.3951
Index build cost	High (LLM calls)	High (LLM calls)	Lookups ($O(E)$)	0.3353	0.8922	0.3514
Graph database	Required	Required	Not needed	0.8922	0.3376	0.3315
Human curation	None	None	Wiki links (excludes)	0.3241	0.8922	0.3315
Query overhead	+graph traversal	+graph traversal	10-1ms	0.3735		
nDCG@10 (MS MARCO)	TBD	TBD				

4.4 Multi-Hop QA Results

Table 5 presents results on multi-hop QA benchmarks. On HotpotQA, Wiki-Link Retrieval achieves statistically significant improvements over the BM25 baseline: nDCG@10 improves from 0.3556 to 0.3735 (+5.0%, $p < 0.001$) with conservative expansion ($\alpha = 0.3$, $m = 1$), while Recall@100 reaches 0.9210 (+22.3%) with larger expansion budgets ($m = 10$). Link coverage analysis shows that 92.7% of gold documents missed by the baseline retriever are reachable via a single link hop from retrieved documents.

On 2WikiMultiHopQA, link expansion improves Recall@100 from 0.5023 to 0.5213 (+3.8%, $p < 0.001$) but decreases nDCG@10 from 0.2171 to 0.1881 (−13.4%). Link coverage analysis shows 90.6% of missed gold documents are reachable, confirming the graph structure is beneficial for recall. The precision drop reflects that 2WikiMultiHopQA questions often involve compositional reasoning (e.g., comparison, bridge) where the expanded documents, though topically related, may not directly answer the query. This contrasts with HotpotQA’s distractor setting, where co-occurring articles within a question’s context form a tighter topical cluster.

Dense retriever results on HotpotQA reveal a contrasting pattern. DPR achieves a strong baseline nDCG@10 of 0.5223—substantially outperforming BM25 (0.3556)—but Wiki-Link Retrieval expansion decreases nDCG@10 to 0.3352 (−35.8%) while only marginally improving R@100 (+3.6%). This mirrors the MS MARCO finding (Table 3): dense retrievers already capture semantic overlap that the link graph provides, so neighbor injection introduces noise into precision-sensitive top ranks. The ColBERT late-interaction model shows the same trend (nDCG@10: 0.0531 $\rightarrow$ 0.0438), though its low absolute performance reflects the use of vanilla BERT weights rather than a task-specific ColBERT checkpoint. These results demonstrate that Wiki-Link Retrieval is particularly effective for datasets where questions require finding multiple linked documents and the supporting facts reside in documents with low lexical overlap to the query but are linked from hub documents that match query terms, primarily benefiting sparse retrievers.

4.5 Comparison with Graph-Augmented RAG

Table 6 compares Wiki-Link Retrieval with graph-augmented RAG approaches across both system complexity and retrieval quality dimensions.

Table 7: Ablation on decay factor α (BM25 as first-stage, $m = 3$, bidirectional). Evaluated on HotpotQA.

α	nDCG@10	Recall@100	MRR@10
0.3	0.3642	0.8922	0.3951
0.5	0.3353	0.8922	0.3514
1.0	0.3241	0.8922	0.3315
0.3735			

Table 8: Ablation on max expansion m ($\alpha = 0.3$, bidirectional). Evaluated on HotpotQA.

m	nDCG@10	Recall@100	MRR@10
1	0.3735	0.8366	0.4255
3	0.3642	0.8922	0.3951
5	0.3524	0.9100	0.3845
10	0.3244	0.9210	0.3809

Table 9: Ablation on link direction ($m = 3$, $\alpha = 0.3$). Evaluated on HotpotQA.

Direction	nDCG@10	Recall@100	MRR@10
Outbound-only	0.3623	0.8890	0.3939
Bidirectional	0.3642	0.8922	0.3951
Baseline (no exp)	0.3556	0.7528	0.4599

Table 10: Ablation on first-stage retriever type ($m = 3$, $\alpha = 0.3$, bidirectional). Evaluated on HotpotQA.

Retriever	Δ nDCG@10	Δ Recall@100
BM25	+0.0179 (+5.0%)	+0.1682 (+22.3%)
DPR	−0.1871 (−35.8%)	+0.0280 (+3.6%)

4.6 Ablation Studies

4.7 Efficiency Analysis

5 DISCUSSION

5.1 When Does Link Expansion Help Most?

Link expansion is most valuable when:

- (1) The relevant document has low lexical overlap with the query but is linked from a keyword-rich hub document
- (2) The corpus has hub/spoke structure with overview pages linking to detailed technical pages
- (3) The query requires multi-hop reasoning — the benchmark results on HotpotQA and 2WikiMultiHopQA confirm this hypothesis

Table 11: Index build and query latency comparison.

Method	Index Build	Query Latency
BM25 (baseline)	$O(n)$ tokenize	$\sim 10\text{ms}$
+ Wiki-Link Retrieval	$+O(E)$ parse	$+ \sim 1\text{ms}$
DPR (baseline)	$O(n)$ embed	$\sim 50\text{ms}$
+ Wiki-Link Retrieval	$+O(E)$ parse	$+ \sim 1\text{ms}$
GraphRAG	$O(n \times \text{LLM})$	$\sim 200\text{ms}$

5.2 Limitations

- (1) Requires existing link structure: only works on corpora with explicit inter-document links. Unlinked or poorly linked corpora will not benefit.
- (2) Expansion budget in dense graphs: in highly connected graphs, most neighbors of a hit are already in the baseline result set, wasting the expansion budget.
- (3) Link quality assumption: not all links are semantically meaningful (e.g., navigational links, boilerplate references). Our method treats all links equally.

5.3 Future Work

- Smart expansion: skip neighbors already in baseline without counting against the expansion budget
- 2-hop expansion: neighbors-of-neighbors with deeper decay ($\alpha^2 = 0.09$)
- Learned link weights: train a lightweight model to predict which links are most useful for expansion
- Hybrid KBs: combining wiki-link expansion with broader document retrieval in mixed-corpora settings

6 CONCLUSION

We presented Wiki-Link Retrieval, a lightweight approach that exploits human-authored link structure in curated knowledge bases for post-retrieval expansion. With zero additional indexing cost (no LLM calls, no graph database) and negligible query overhead ($\sim 1\text{ms}$), it improves retrieval quality across standard IR benchmarks and is particularly effective for multi-hop queries. Our approach fills a gap in the current RAG landscape where no mainstream system exploits explicit inter-document links at the retrieval level, and achieves competitive quality with graph-augmented RAG approaches at a fraction of the cost.

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